Mapping Lake Temperature Model

Purpose

- Find and identify patterns in how changing the constants of elmtest.exe, an environmental model, affects its outputs.

What is elmtest.exe?

- Fortran binary
- Compiled in a docker image from [Docker4SPEL]
- Environmental model

elmtest.exe interface

- Inputs: lakeparams.txt
- Outputs: cpu_LakeTemperatureTest.txt

lakeparams.txt

CONTAINER:/app/elmtest/lakeparams.txt

```
betavis
1.0
za_lake
2.0
n2min
7.5e-05
tdmax
275.0
pudz
11.0
mixfact
0.5
depthcrit
25.0
```

cpu_LakeTemperatureTest.txt

CONTAINER:/app/elmtest/cpu_LakeTemperatureTest.txt (Default Parameters)

```
col_ws%h2osno
    0.000000000000000          0.000000000000000          0.000000000000000
    0.000000000000000          0.000000000000000          0.000000000000000
    0.000000000000000          0.000000000000000          0.000000000000000
    0.000000000000000          0.000000000000000          0.000000000000000
    0.000000000000000          0.000000000000000          0.000000000000000
    0.000000000000000          0.000000000000000          0.000000000000000
                                      - - - … - - -
col_es%hc_soi
  1.000000000000000E+036   1.000000000000000E+036   1.000000000000000E+036
  1.000000000000000E+036   1.000000000000000E+036   1.000000000000000E+036
  1.000000000000000E+036   1.000000000000000E+036   1.000000000000000E+036
  1.000000000000000E+036   1.000000000000000E+036   1.000000000000000E+036
  1.000000000000000E+036   1.000000000000000E+036   1.000000000000000E+036
  1.000000000000000E+036    15270.67984628235       1.000000000000000E+036
                                      - - - … - - -
```

Approach

- Write change to lakeparams.txt
- Run elmtest.exe
- Compare cpu_LakeTemperatureTest.txt to cpu_LakeTemperatureRef.txt
- Repeat

Interface of mapping script

- Orders
  - Represent a linear sequence of values for a specified parameter
- Output
  - Console output and/or csv output

Example Order: betavis_min_to_max.json

lake_temp_work/lake_temperature_mapper/mapping_orders/betavis_min_to_max.json

```json
{
  "param":"betavis",
  "samples":11,
  "start":"m",
  "end":"M"
}
```

Results in: [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]

Example Console Output

CONTAINER:/app/scripts/cli.py running betavis_min_to_max order.

```
Executing order: betavis_min_to_max
betavis: 0.0 -> 1.0, 11 samples

Set betavis to 0.0
lakestate_vars%betaprime_col
1.0                    -> 0.0                    by -1.0                  at 16
1.0                    -> 0.0                    by -1.0                  at 33
1.0                    -> 0.0                    by -1.0                  at 50
                                    - - - … - - -
Set betavis to 0.1
lakestate_vars%betaprime_col
1.0                    -> 0.1                    by -0.9                  at 16
1.0                    -> 0.1                    by -0.9                  at 33
1.0                    -> 0.1                    by -0.9                  at 50
                                    - - - … - - -
Set betavis to 1.0
No differences
```

Example CSV Output

CONTAINER:/app/scripts/cli.py running betavis_min_to_max order with -s flag.
CONTAINER:/app/mapping_output/betavis_min_to_max.csv

```
betavis,0.0
lakestate_vars%betaprime_col
1.0,0.0,-1.0,16
1.0,0.0,-1.0,33
1.0,0.0,-1.0,50
                                        - - - … - - -

betavis,0.1
lakestate_vars%betaprime_col
1.0,0.1,-0.9,16
1.0,0.1,-0.9,33
1.0,0.1,-0.9,50
                                        - - - … - - -

betavis,1.0
```

Development using fake_binary.exe

- The Docker container is stateless
- elmtest.exe depends on the environment inside the docker image
- fake_binary.exe was developed to imitate elmtest.exe but without depending on the docker image
  - Uses known functions to compute output, making testing easier

fake_LakeTemperatureTest.txt

lake_temp_work/lake_temperature_mapper/fake_binary/fake_LakeTemperatureTest.txt
(Default Parameters)

```
fakevar%sum
  314.500075 629.000150
  31.450007 157.250037
fakevar%product
  31.450007 157.250037
  5.671875 32.170166
```

Example Console Output with fake_binary

lake_temp_work/lake_temperature_mapper/scripts/cli.py running betavis_min_to_max order.

```
Executing order: betavis_min_to_max
betavis: 0.0 -> 1.0, 11 samples

Set betavis to 0.0
fakevar%sum
314.500075                -> 313.500075                by -1.0                   at 0
629.00015                 -> 627.00015                 by -2.0                   at 1
31.450007                 -> 31.350008                 by -0.0999990000000004    at 2
                                              - - - … - - -

Set betavis to 0.1
fakevar%sum
314.500075                -> 313.600075                by -0.8999999999999773    at 0
629.00015                 -> 627.20015                 by -1.7999999999999545    at 1
31.450007                 -> 31.360008                 by -0.08999899999999883   at 2
                                              - - - … - - -

Set betavis to 1.0
No differences
```

Configuration of cli.py

- Configuration files necessary to account for differences between naming of elmtest.exe and fake_binary.exe files.
- Can generate a configuration file using configure_mapper.py

Configuration files

lake_temp_work/lake_temperature_mapper/config/mapper.conf

```
binary_path: fake_binary/fake_binary.exe
range_path: fake_binary/FUT_lake_range.txt
params_path: fake_binary/lakeparams.txt
defaults_path: fake_binary/lakeparams_defaults.txt
ref_output: fake_binary/fake_LakeTemperatureRef.txt
test_output: fake_binary/fake_LakeTemperatureTest.txt
order_directory: mapping_orders
output_directory: mapping_output
binary_args:
```

CONTAINER:/app/config/mapper.conf

```
binary_path: elmtest/elmtest.exe
range_path: elmtest/FUT_lake_range.txt
params_path: elmtest/lakeparams.txt
defaults_path: elmtest/lakeparams_defaults.txt
ref_output: elmtest/cpu_LakeTemperatureRef.txt
test_output: elmtest/cpu_LakeTemperatureTest.txt
order_directory: mapping_orders
output_directory: mapping_output
binary_args: 1
```